

# Enterprise Database Investigation s Architecture Reflection

## Scenario

You have recently joined a software company as part of a technical engineering team.

The company is planning to build large-scale systems for different industries such as banking, healthcare, e-commerce, education, and government sectors.

Before starting development, the architecture team must evaluate different Database Management Systems (DBMSs) to make the correct long-term technology decision.

Your task is to investigate and reflect on how companies choose database technologies in real industry environments.

This is a full-day investigation and reflection task designed to introduce you to real-world database engineering concepts beyond classroom-level SQL syntax.

## Objective

This task is designed to help you understand that database systems are not only about writing SQL queries.

Modern companies choose database technologies based on:

- business requirements,
- scalability,
- security,
- ecosystem integration,
- enterprise support,
- cloud infrastructure,
- licensing,
- cost,
- and long-term system architecture decisions.

The goal is to begin thinking like a software engineer and system architect rather than only focusing on syntax.

## **Investigation Areas**

### **Part 1 – DBMS Technologies Investigation**

Investigate the following DBMS technologies:

- Oracle
- SQL Server
- PostgreSQL
- MySQL

For each technology, investigate:

- Common industries using it
- Typical company/system environments
- Enterprise vs open-source usage
- Strengths and trade-offs
- Common tools associated with it
- Integration with cloud or enterprise systems
- Licensing/commercial considerations

### **Part 2 – Enterprise Decision Thinking**

Reflect on the following:

- Why might a bank choose Oracle instead of MySQL?
- Why might a startup choose PostgreSQL instead of Oracle?
- Why do some companies stay with Microsoft technologies such as SQL Server?
- How does cloud infrastructure affect database decisions?
- Why do enterprise companies sometimes pay for commercial database systems instead of using free alternatives?

## Part 3 – Architecture s Scalability Thinking

Investigate how different DBMS systems may differ in:

- scalability,
- performance,
- security,
- licensing,
- enterprise support,
- cloud readiness,
- backup/recovery strategies,
- and integration with other technologies.

Additionally investigate:

- Which DBMS systems are commonly used in:
  - enterprise systems,
  - startups,
  - government sectors,
  - cloud-native systems,
  - banking systems,
  - and large transactional environments.

## Part 4 – Migration s Industry Reflection

Reflect carefully on the following question:

If most DBMS systems support:

- tables,
- relationships,
- joins,
- procedures,
- triggers,
- and transactions,

Then why do companies still choose different DBMS technologies instead of using one universal system?

Additionally reflect on:

- Why companies migrate from one DBMS to another
- What technical and business challenges may appear during migration
- Why migration projects are considered high-risk in enterprise environments
- How database architecture decisions affect long-term software systems

## **Final Deliverable**

Prepare a professional engineering reflection document summarizing your investigation and analysis.

Your submission should focus on:

- analytical thinking,
- architecture awareness,
- enterprise understanding,
- professional technical reasoning,
- and system-level thinking.

The purpose of this task is not to memorize definitions, but to begin understanding how real-world technology decisions are made in professional software engineering environments.

## **Expected Outcome**

By the end of this task, you should begin understanding:

- Why do multiple DBMS technologies exist,
- Why companies make different architecture decisions,
- how enterprise systems differ from traditional classroom projects,
- and why strong engineers focus on concepts and architecture rather than depending on a single technology or syntax.